

Shaun K. Kane

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h-index 49, 10,767 citations (Google Scholar, June 2026)

Employment

Google, Boulder, CO

- Senior Staff Research Scientist, Human-Centered AI & Foundational Machine Learning (2026–)
- Staff Research Scientist, Human-Centered AI & Foundational Machine Learning (2022–2026)

University of Colorado Boulder, Boulder, CO

- Associate Professor Adjunct (2024–)
- Associate Professor, Department of Computer Science (2018–2024)
- Assistant Professor, Department of Computer Science (2014–2018)

University of Maryland Baltimore County, Baltimore, MD

- Assistant Professor, Department of Information Systems (2011–2014)

Education

- Ph.D. in Information Science, University of Washington, Seattle WA (2005–2011). *Co-Chairs: Jacob O. Wobbrock and Richard E. Ladner.*
- M.S. in Information Science, University of Washington, Seattle WA (2011)
- M.S. in Computer Science, University of Massachusetts, Amherst MA (2003–2005)
- B.S. in Computer Science, University of Massachusetts, Amherst MA (1999–2003)

Publications

Preprints / Manuscripts Under Review

- [Pr.1] A. Olmos, A. K. Sinha, R. D. Santos, R. R. Rodriguez, J. A. Landay, S. S. Sepah, P. Nelson, and S. K. Kane. Making Videos Accessible for Blind and Low Vision Users Using a Multimodal Agent Video Player. 2026. DOI: [10.48550/ARXIV.2602.04104](#).

Journal Articles

- [J.12] C. L. Bennett, A. Pavel, A. Bryant, L. J. McCarthy Eberly, R. Shelby, and S. K. Kane. “Made by People, Described by People”: The Changing Work Practices of Audio Description Professionals. In: *ACM Transactions on Accessible Computing* 18.4 (Dec. 2025), pp. 1–28. DOI: [10.1145/3777475](#).
- [J.11] S. Cai, S. Venugopalan, K. Seaver, X. Xiao, K. Tomanek, S. Jalsutram, M. R. Morris, S. Kane, A. Narayanan, R. L. MacDonald, E. Kornman, D. Vance, B. Casey, S. M. Gleason, P. Q. Nelson, and M. P. Brenner. Using large language models to accelerate communication for eye gaze typing users with ALS. In: *Nature Communications* 15.1 (Nov. 2024). DOI: [10.1038/s41467-024-53873-3](#).
- [J.10] P. Parekh, J. L. Polman, S. Kane, and R. B. Shapiro. Reconfiguring science education through caring human inquiry and design with pets. In: *Journal of the Learning Sciences* 32.4-5 (Aug. 2023), pp. 487–533. DOI: [10.1080/10508406.2023.2228990](#).
- [J.9] V. Koushik and S. K. Kane. Towards augmented reality coaching for daily routines: Participatory design with individuals with cognitive disabilities and their caregivers. In: *International Journal of Human-Computer Studies* 165 (Sept. 2022), p. 102862. DOI: [10.1016/j.ijhcs.2022.102862](#).
- [J.8] A. Muehlbradt and S. K. Kane. What’s in an ALT Tag? Exploring Caption Content Priorities through Collaborative Captioning. In: *ACM Transactions on Accessible Computing* 15.1 (Mar. 2022), pp. 1–32. DOI: [10.1145/3507659](#).

- [J.7] H. P. Profita, A. Stangl, L. Matuszewska, S. Sky, R. Kushalnagar, and S. K. Kane. “Wear It Loud”: How and Why Hearing Aid and Cochlear Implant Users Customize Their Devices. In: *ACM Transactions on Accessible Computing* 11.3 (Sept. 2018), pp. 1–32. DOI: [10.1145/3214382](https://doi.org/10.1145/3214382).
- [J.6] J. O. Wobbrock, K. Z. Gajos, S. K. Kane, and G. C. Vanderheiden. Ability-based design. In: *Communications of the ACM* 61.6 (May 2018), pp. 62–71. DOI: [10.1145/3148051](https://doi.org/10.1145/3148051).
- [J.5] H. Profita, M. Lightner, N. Correll, and S. Kane. Textile-based wearables. In: *Journal on Technology and Persons with Disabilities*, 5, pp. 40-50 (2017).
- [J.4] P. Carrington, J.-M. Chang, K. Chang, C. Hornback, A. Hurst, and S. K. Kane. The Gest-Rest Family: Exploring Input Possibilities for Wheelchair Armrests. In: *ACM Transactions on Accessible Computing* 8.3 (Apr. 2016), pp. 1–24. DOI: [10.1145/2873062](https://doi.org/10.1145/2873062).
- [J.3] U. Oh, S. Branham, L. Findlater, and S. K. Kane. Audio-Based Feedback Techniques for Teaching Touchscreen Gestures. In: *ACM Transactions on Accessible Computing* 7.3 (Nov. 2015), pp. 1–29. DOI: [10.1145/2764917](https://doi.org/10.1145/2764917).
- [J.2] J. O. Wobbrock, S. K. Kane, K. Z. Gajos, S. Harada, and J. Froehlich. Ability-Based Design: Concept, Principles and Examples. In: *ACM Transactions on Accessible Computing* 3.3 (Apr. 2011), pp. 1–27. DOI: [10.1145/1952383.1952384](https://doi.org/10.1145/1952383.1952384).
- [J.1] S. Kane. Everyday inclusive web design: an activity perspective. In: *Information Research*, 12 (1), 16 pages (2007).

Refereed Conference Papers

- [C.79] F. M. Li, M. X. Liu, C. L. Bennett, and S. K. Kane. ADCanvas: Accessible and Conversational Audio Description Authoring for Blind and Low Vision Creators. In: *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*. CHI '26. ACM, Apr. 2026, pp. 1–21. DOI: [10.1145/3772318.3791158](https://doi.org/10.1145/3772318.3791158).
- [C.78] C. L. Mitchell, J. Kong, J. J. Martinez, S. K. Kane, A. J. Ko, A. Hiniker, and J. O. Wobbrock. Ability Heuristics for Conducting Accessibility Inspections. In: *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*. CHI '26. ACM, Apr. 2026, pp. 1–15. DOI: [10.1145/3772318.3791504](https://doi.org/10.1145/3772318.3791504).
- [C.77] C. L. Bennett, S. K. Kane, and C. N. Harrington. Toward Community-Led Evaluations of Text-to-Image AI Representations of Disability, Health, and Accessibility. In: *Proceedings of the 5th ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization*. EAAMO '25. ACM, Nov. 2025, pp. 256–270. DOI: [10.1145/3757887.3763012](https://doi.org/10.1145/3757887.3763012).
- [C.76] J. E. Froehlich, A. J. Fiannaca, N. M. Jaber, V. Tsaran, and S. K. Kane. Making Street View Accessible Using Context-Aware, Multimodal AI: A Demo of StreetReaderAI. In: *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*. ASSETS '25. ACM, Oct. 2025, pp. 1–6. DOI: [10.1145/3663547.3759762](https://doi.org/10.1145/3663547.3759762).
- [C.75] S. Moharana, C. L. Bennett, E. Buehler, M. Madaio, V. Tibdewal, and S. K. Kane. “Accessibility People, You Go Work on That Thing of Yours over There”: Addressing Disability Inclusion in AI Product Organizations. In: *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society* 8.2 (Oct. 2025), pp. 1724–1737. DOI: [10.1609/aies.v8i2.36669](https://doi.org/10.1609/aies.v8i2.36669).
- [C.72] C. L. Bennett, R. Shelby, N. Rostamzadeh, and S. K. Kane. Painting with Cameras and Drawing with Text: AI Use in Accessible Creativity. In: *The 26th International ACM SIGACCESS Conference on Computers and Accessibility*. ASSETS '24. ACM, Oct. 2024, pp. 1–19. DOI: [10.1145/3663548.3675644](https://doi.org/10.1145/3663548.3675644).

- [C.69] M. Das, A. J. Fiannaca, M. R. Morris, S. K. Kane, and C. L. Bennett. From Provenance to Aberrations: Image Creator and Screen Reader User Perspectives on Alt Text for AI-Generated Images. In: *Proceedings of the CHI Conference on Human Factors in Computing Systems*. CHI '24. ACM, May 2024, pp. 1–21. DOI: [10.1145/3613904.3642325](https://doi.org/10.1145/3613904.3642325).
- [C.73] V. Gadiraju, L. Jayne, and S. K. Kane. “It’s an independent living skill, but covered with fun!”: Prompting At-Home Skill Development for Children with Vision Impairment. In: *The 26th International ACM SIGACCESS Conference on Computers and Accessibility*. ASSETS '24. ACM, Oct. 2024, pp. 1–14. DOI: [10.1145/3663548.3675626](https://doi.org/10.1145/3663548.3675626).
- [C.70] F. M. Li, M. X. Liu, S. K. Kane, and P. Carrington. A Contextual Inquiry of People with Vision Impairments in Cooking. In: *Proceedings of the CHI Conference on Human Factors in Computing Systems*. CHI '24. ACM, May 2024, pp. 1–14. DOI: [10.1145/3613904.3642233](https://doi.org/10.1145/3613904.3642233).
- [C.74] F. M. Li, A. Wang, P. Carrington, and S. K. Kane. A Recipe for Success? Exploring Strategies for Improving Non-Visual Access to Cooking Instructions. In: *The 26th International ACM SIGACCESS Conference on Computers and Accessibility*. ASSETS '24. ACM, Oct. 2024, pp. 1–15. DOI: [10.1145/3663548.3675662](https://doi.org/10.1145/3663548.3675662).
- [C.71] K. A. Mack, R. Qadri, R. Denton, S. K. Kane, and C. L. Bennett. “They only care to show us the wheelchair”: disability representation in text-to-image AI models. In: *Proceedings of the CHI Conference on Human Factors in Computing Systems*. CHI '24. ACM, May 2024, pp. 1–23. DOI: [10.1145/3613904.3642166](https://doi.org/10.1145/3613904.3642166).
- [C.67] M. Bowman, J. Robinson, E. Buehler, and S. Kane. “I just thought it was me”: How Smartphones Fail Users with Mild-to-Moderate Dexterity Differences. In: *The 25th International ACM SIGACCESS Conference on Computers and Accessibility*. ASSETS '23. ACM, Oct. 2023, pp. 1–12. DOI: [10.1145/3597638.3608396](https://doi.org/10.1145/3597638.3608396).
- [C.64] S. Cai, S. Venugopalan, K. Tomanek, S. Kane, M. R. Morris, R. Cave, R. Macdonald, J. Campbell, B. Casey, E. Kornman, D. E. Vance, and J. Beavers. SpeakFaster Observer: Long-Term Instrumentation of Eye-Gaze Typing for Measuring AAC Communication. In: *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. CHI '23. ACM, Apr. 2023, pp. 1–8. DOI: [10.1145/3544549.3573870](https://doi.org/10.1145/3544549.3573870).
- [C.68] V. Gadiraju, S. Kane, S. Dev, A. Taylor, D. Wang, R. Denton, and R. Brewer. “I wouldn’t say offensive but...”: Disability-Centered Perspectives on Large Language Models. In: *2023 ACM Conference on Fairness Accountability and Transparency*. FAccT '23. ACM, June 2023, pp. 205–216. DOI: [10.1145/3593013.3593989](https://doi.org/10.1145/3593013.3593989).
- [C.63] V. Koushik and S. Kane. Ability + Motivation: Understanding Factors that Influence People with Cognitive Disabilities in Regularly Practicing Daily Activities. In: *20th International Web for All Conference*. W4A '23. ACM, Apr. 2023, pp. 122–133. DOI: [10.1145/3587281.3587295](https://doi.org/10.1145/3587281.3587295).
- [C.66] S. Valencia, R. Cave, K. Kallarackal, K. Seaver, M. Terry, and S. K. Kane. “The less I type, the better”: How AI Language Models can Enhance or Impede Communication for AAC Users. In: *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. CHI '23. ACM, Apr. 2023, pp. 1–14. DOI: [10.1145/3544548.3581560](https://doi.org/10.1145/3544548.3581560).
- [C.65] Q. Wang, M. Madaio, S. Kane, S. Kapania, M. Terry, and L. Wilcox. Designing Responsible AI: Adaptations of UX Practice to Meet Responsible AI Challenges. In: *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. CHI '23. ACM, Apr. 2023, pp. 1–16. DOI: [10.1145/3544548.3581278](https://doi.org/10.1145/3544548.3581278).

- [C.62] A. Muehlbradt, G. Whiting, S. Kane, and L. Devendorf. Knitting Access: Exploring Stateful Textiles with People with Disabilities. In: *Designing Interactive Systems Conference*. DIS '22. ACM, June 2022, pp. 1058–1070. DOI: [10.1145/3532106.3533551](https://doi.org/10.1145/3532106.3533551).
- [C.61] P. Parekh, J. Polman, S. Kane, and R. Shapiro. Learning science while caring for pets: Findings from a virtual workshop for teens. In: *Proceedings of ICLS '22*, 321–328. 2022.
- [C.60] P. Parekh, J. Polman, S. Kane, and R. Shapiro. Teens and pets as participatory design partners. In: *Proceedings of ICLS '22*, 386–389. 2022.
- [C.58] V. Gadiraju, O. Doyle, and S. Kane. Exploring Technology Design for Students with Vision Impairment in the Classroom and Remotely. In: *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. CHI '21. ACM, May 2021, pp. 1–13. DOI: [10.1145/3411764.3445755](https://doi.org/10.1145/3411764.3445755).
- [C.59] V. Gadiraju, J. Garcia, S. Kane, and A. M. Brock. “It is fascinating to make these beasts fly”: Understanding Visually Impaired People’s Motivations and Needs for Drone Piloting. In: *Proceedings of the 23rd International ACM SIGACCESS Conference on Computers and Accessibility*. ASSETS '21. ACM, Oct. 2021, pp. 1–12. DOI: [10.1145/3441852.3471219](https://doi.org/10.1145/3441852.3471219).
- [C.54] V. Gadiraju, A. Muehlbradt, and S. K. Kane. BrailleBlocks: Computational Braille Toys for Collaborative Learning. In: *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. CHI '20. ACM, Apr. 2020, pp. 1–12. DOI: [10.1145/3313831.3376295](https://doi.org/10.1145/3313831.3376295).
- [C.53] G. M. Johnson and S. K. Kane. Game changer: accessible audio and tactile guidance for board and card games. In: *Proceedings of the 17th International Web for All Conference*. W4A '20. ACM, Apr. 2020, pp. 1–12. DOI: [10.1145/3371300.3383347](https://doi.org/10.1145/3371300.3383347).
- [C.57] S. K. Kane, A. Guo, and M. R. Morris. Sense and Accessibility: Understanding People with Physical Disabilities’ Experiences with Sensing Systems. In: *Proceedings of the 22nd International ACM SIGACCESS Conference on Computers and Accessibility*. ASSETS '20. ACM, Oct. 2020, pp. 1–14. DOI: [10.1145/3373625.3416990](https://doi.org/10.1145/3373625.3416990).
- [C.55] A. Kelly, C. Chang, C. Hill, M. West, M. Yoder, J. Polman, S. Kane, M. Eisenberg, and R. Shapiro. Our dog probably thinks Christmas is really boring”: Re-mediating science education for feminist-inspired inquiry. In: *Proceedings of ICLS '20*, 935–942. 2020.
- [C.56] M. Mott, J. Tang, S. Kane, E. Cutrell, and M. Ringel Morris. “I just went into it assuming that I wouldn’t be able to have the full experience”: Understanding the Accessibility of Virtual Reality for People with Limited Mobility. In: *Proceedings of the 22nd International ACM SIGACCESS Conference on Computers and Accessibility*. ASSETS '20. ACM, Oct. 2020, pp. 1–13. DOI: [10.1145/3373625.3416998](https://doi.org/10.1145/3373625.3416998).
- [C.52] D. Guinness, A. Muehlbradt, D. Szafir, and S. K. Kane. RoboGraphics: Using Mobile Robots to Create Dynamic Tactile Graphics. In: *Proceedings of the 21st International ACM SIGACCESS Conference on Computers and Accessibility*. ASSETS '19. ACM, Oct. 2019, pp. 673–675. DOI: [10.1145/3308561.3354597](https://doi.org/10.1145/3308561.3354597).
- [C.48] V. Koushik, D. Guinness, and S. K. Kane. StoryBlocks: A Tangible Programming Game To Create Accessible Audio Stories. In: *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. CHI '19. ACM, May 2019, pp. 1–12. DOI: [10.1145/3290605.3300722](https://doi.org/10.1145/3290605.3300722).
- [C.49] V. Koushik and S. K. Kane. “It Broadens My Mind”: Empowering People with Cognitive Disabilities through Computing Education. In: *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. CHI '19. ACM, May 2019, pp. 1–12. DOI: [10.1145/3290605.3300744](https://doi.org/10.1145/3290605.3300744).
- [C.50] K. Reinholt, D. Guinness, and S. K. Kane. EyeDescribe: Combining Eye Gaze and Speech to Automatically Create Accessible Touch Screen Artwork. In: *Proceedings of the 2019 ACM Interna-*

tional Conference on Interactive Surfaces and Spaces. ISS '19. ACM, Nov. 2019, pp. 101–112. DOI: [10.1145/3343055.3359722](https://doi.org/10.1145/3343055.3359722).

- [C.51] A. Zimmermann-Niefeld, M. Turner, B. Murphy, S. K. Kane, and R. B. Shapiro. Youth Learning Machine Learning through Building Models of Athletic Moves. In: *Proceedings of the 18th ACM International Conference on Interaction Design and Children*. IDC '19. ACM, June 2019, pp. 121–132. DOI: [10.1145/3311927.3323139](https://doi.org/10.1145/3311927.3323139).
- [C.45] P. K. Chilana, N. Hudson, S. Bhaduri, P. Shashikumar, and S. Kane. Supporting Remote Real-Time Expert Help: Opportunities and Challenges for Novice 3D Modelers. In: *2018 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*. IEEE, Oct. 2018, pp. 157–166. DOI: [10.1109/vlhcc.2018.8506568](https://doi.org/10.1109/vlhcc.2018.8506568).
- [C.47] D. Guinness, A. Muehlbradt, D. Szafer, and S. K. Kane. The Haptic Video Player: Using Mobile Robots to Create Tangible Video Annotations. In: *Proceedings of the 2018 ACM International Conference on Interactive Surfaces and Spaces*. ISS '18. ACM, Nov. 2018, pp. 203–211. DOI: [10.1145/3279778.3279805](https://doi.org/10.1145/3279778.3279805).
- [C.44] S. K. Kane, V. Koushik, and A. Muehlbradt. Bonk: accessible programming for accessible audio games. In: *Proceedings of the 17th ACM Conference on Interaction Design and Children*. IDC '18. ACM, June 2018, pp. 132–142. DOI: [10.1145/3202185.3202754](https://doi.org/10.1145/3202185.3202754).
- [C.46] A. Muehlbradt, M. Atreya, D. Guinness, and S. K. Kane. Exploring the Design of Audio-Kinetic Graphics for Education. In: *Proceedings of the 20th ACM International Conference on Multimodal Interaction*. ICMI '18. ACM, Oct. 2018, pp. 455–463. DOI: [10.1145/3242969.3243004](https://doi.org/10.1145/3242969.3243004).
- [C.43] M. Whitlock, E. Harnner, J. R. Brubaker, S. Kane, and D. A. Szafer. Interacting with Distant Objects in Augmented Reality. In: *2018 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. IEEE, Mar. 2018, pp. 41–48. DOI: [10.1109/vr.2018.8446381](https://doi.org/10.1109/vr.2018.8446381).
- [C.42] S. Bhaduri, J. G. Ortiz Tovar, and S. K. Kane. Fabrication Games: Using 3D Printers to Explore New Interactions for Tabletop Games. In: *Proceedings of the 2017 ACM SIGCHI Conference on Creativity and Cognition*. C&C '17. ACM, June 2017, pp. 51–62. DOI: [10.1145/3059454.3059463](https://doi.org/10.1145/3059454.3059463).
- [C.38] L. E. Boyd, K. Rector, H. Profita, A. J. Stangl, A. Zolyomi, S. K. Kane, and G. R. Hayes. Understanding the Role Fluidity of Stakeholders During Assistive Technology Research “In the Wild”. In: *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. CHI '17. ACM, May 2017, pp. 6147–6158. DOI: [10.1145/3025453.3025493](https://doi.org/10.1145/3025453.3025493).
- [C.39] A. M. Feit, S. Williams, A. Toledo, A. Paradiso, H. Kulkarni, S. Kane, and M. R. Morris. Toward Everyday Gaze Input: Accuracy and Precision of Eye Tracking and Implications for Design. In: *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. CHI '17. ACM, May 2017, pp. 1118–1130. DOI: [10.1145/3025453.3025599](https://doi.org/10.1145/3025453.3025599).
- [C.40] D. Guinness, D. Szafer, and S. K. Kane. GUI Robots: Using Off-the-Shelf Robots as Tangible Input and Output Devices for Unmodified GUI Applications. In: *Proceedings of the 2017 Conference on Designing Interactive Systems*. DIS '17. ACM, June 2017, pp. 767–778. DOI: [10.1145/3064663.3064706](https://doi.org/10.1145/3064663.3064706).
- [C.41] S. K. Kane and M. R. Morris. Let’s Talk About X: Combining Image Recognition and Eye Gaze to Support Conversation for People with ALS. In: *Proceedings of the 2017 Conference on Designing Interactive Systems*. DIS '17. ACM, June 2017, pp. 129–134. DOI: [10.1145/3064663.3064762](https://doi.org/10.1145/3064663.3064762).
- [C.37] S. K. Kane, M. R. Morris, A. Paradiso, and J. Campbell. “At times avuncular and cantankerous, with the reflexes of a mongoose”: Understanding Self-Expression through Augmentative and Alternative Communication Devices. In: *Proceedings of the 2017 ACM Conference on Computer Supported*

Cooperative Work and Social Computing. CSCW '17. ACM, Feb. 2017, pp. 1166–1179. DOI: [10.1145/2998181.2998284](https://doi.org/10.1145/2998181.2998284).

- [C.33] M. R. Morris, A. Zolyomi, C. Yao, S. Bahram, J. P. Bigham, and S. K. Kane. “With most of it being pictures now, I rarely use it”: Understanding Twitter’s Evolving Accessibility to Blind Users. In: *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. CHI '16. ACM, May 2016, pp. 5506–5516. DOI: [10.1145/2858036.2858116](https://doi.org/10.1145/2858036.2858116).
- [C.35] M. E. Mott, R.-D. Vatavu, S. K. Kane, and J. O. Wobbrock. Smart Touch: Improving Touch Accuracy for People with Motor Impairments with Template Matching. In: *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. CHI '16. ACM, May 2016, pp. 1934–1946. DOI: [10.1145/2858036.2858390](https://doi.org/10.1145/2858036.2858390).
- [C.34] H. Profita, R. Albaghli, L. Findlater, P. Jaeger, and S. K. Kane. The AT Effect: How Disability Affects the Perceived Social Acceptability of Head-Mounted Display Use. In: *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. CHI '16. ACM, May 2016, pp. 4884–4895. DOI: [10.1145/2858036.2858130](https://doi.org/10.1145/2858036.2858130).
- [C.36] H. P. Profita, A. Stangl, L. Matuszewska, S. Sky, and S. K. Kane. Nothing to Hide: Aesthetic Customization of Hearing Aids and Cochlear Implants in an Online Community. In: *Proceedings of the 18th International ACM SIGACCESS Conference on Computers and Accessibility*. ASSETS '16. ACM, Oct. 2016, pp. 219–227. DOI: [10.1145/2982142.2982159](https://doi.org/10.1145/2982142.2982159).
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- [C.30] P. Carrington, S. Hosmer, T. Yeh, A. Hurst, and S. Kane. Like this, but better”: supporting novices’ design and fabrication of 3D models using existing objects. In: *Proceedings of iConference '15*. (acceptance rate 36%). 2015.
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Refereed Posters, Extended Abstracts, and Presentations

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- [P.19] S. Kane, H. Profita, M. Lightner, and N. Correll. Assistive wearables: emerging trends and design considerations. In: *Refereed presentation at CSUN '17 Conference*. 2017.
- [P.20] V. Koushik and S. K. Kane. Tangibles + Programming + Audio Stories = Fun. In: *Proceedings of the 19th International ACM SIGACCESS Conference on Computers and Accessibility*. ASSETS '17. ACM, Oct. 2017, pp. 341–342. DOI: [10.1145/3132525.3134769](https://doi.org/10.1145/3132525.3134769).
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- [P.14] S. Kane and S. Bahram. Designing 3D-printed tactile comic books. In: *Refereed presentation at CSUN '15 Conference*. 2015.
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Workshop Papers

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- [W.4] E. Kreiss, S. Venugopalan, S. Kane, and M. Morris. Practical challenges for investigating abbreviation strategies. In: *CHI 2023 Workshop on Assistive Writing, 3 pages*. 2023.

- [W.5] A. Taheri, M. Izadi, G. Shriram, N. Rostamzadeh, and S. Kane. Breaking Barriers to Creative Expression: Co-Designing and Implementing an Accessible Text-to-Image Interface. In: *Workshop on Machine Learning for Creativity and Design at NeurIPS '23*. 2023.
- [W.3] A. Fiannaca, C. Bennett, S. Kane, and M. Morris. Beyond Safety: toward a value-sensitive approach to the design of AI systems. In: *NeurIPS 2022 Human-Centered AI Workshop*, 6 pages. 2022.
- [W.1] M. Williams, A. Hurst, and S. Kane. Are you going to help her?" – ethical dilemmas presented by research with blind participants. In: *Proceedings of CSCW '14*, 4 pages. 2014.
- [W.2] A. Hurst and S. Kane. Making "making" accessible. In: *Proceedings of the 12th International Conference on Interaction Design and Children*. IDC '13. ACM, June 2013, pp. 635–638. DOI: [10.1145/2485760.2485883](https://doi.org/10.1145/2485760.2485883).

Book Chapters

- [BC.4] S. Kane. Mobile Development + Accessibility. In: *In Alannah Oleson, Amy J. Ko, and Richard E. Ladner (Eds.). Teaching Accessible Computing*. 2025.
- [BC.3] S. K. Kane. Wearables. In: *Web Accessibility*. Springer London, 2019, pp. 701–714. DOI: [10.1007/978-1-4471-7440-0_35](https://doi.org/10.1007/978-1-4471-7440-0_35).
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- [BC.1] S. Kane, J. Hannah, P. Edwards, and J. Dorman. Teaching in computer classrooms. In: *In C. Ross and J. Dunphy (Eds.), Strategies for Teaching Assistant and International Teaching Assistant Development: Beyond Micro Teaching*. Bolton, MA: Anker, 48-52. 2007.

Invited Articles

- [IA.4] A. Zimmermann-Niefield, R. B. Shapiro, and S. Kane. Sports and machine learning: How young people can use data from their own bodies to learn about machine learning. In: *XRDS: Crossroads, The ACM Magazine for Students* 25.4 (July 2019), pp. 44–49. DOI: [10.1145/3331071](https://doi.org/10.1145/3331071).
- [IA.3] S. Kane. Invisibility, conspicuousness, and accessibility: how sensing systems fail for non-traditional users. In: *Proceedings of HCIC '16*, 3 pages (2016).
- [IA.2] S. K. Kane, A. Hurst, E. Buehler, P. A. Carrington, and M. A. Williams. Collaboratively designing assistive technology. In: *Interactions* 21.2 (Mar. 2014), pp. 78–81. DOI: [10.1145/2566462](https://doi.org/10.1145/2566462).
- [IA.1] S. K. Kane. Context-enhanced interaction techniques for more accessible mobile phones. In: *ACM SIGACCESS Accessibility and Computing* 93 (Jan. 2009), pp. 39–43. DOI: [10.1145/1531930.1531936](https://doi.org/10.1145/1531930.1531936).

Invited Presentations

- [IP.16] J. Bigham, W. Lasecki, and S. Kane. AI for accessibility: augmenting sensory capabilities with intelligent technology. Microsoft Research Faculty Summit. 2017.
- [IP.15] S. Kane. Our assistive technologies, ourselves: understanding the relationships between wearable technology, accessibility, and identity. University of Texas Information School. 2017.
- [IP.12] S. Kane. Combining ability and context. Microsoft Research. 2016.
- [IP.13] S. Kane. Combining ability and context to create more accessible user interfaces. University of Washington DUB Seminar. 2016.

- [IP.14] S. Kane. Towards superhuman computing. Commencement speech, University of Washington Information School. 2016.
- [IP.11] S. Kane. Superhuman computing: designing custom software and hardware interfaces to support our natural abilities. Keynote talk at 2015 Richard Tapia Conference on Diversity and Computing. 2015.
- [IP.10] S. Kane. Superhuman computing: designing technology to support our natural abilities. University of Colorado ECSITE Program. 2015.
- [IP.9] S. Kane. Gesture-based user interfaces for a more accessible world. Human-Computer Interaction Institute, Carnegie Mellon University. 2013.
- [IP.7] S. Kane. Mobile gesture-based user interfaces for people with disabilities. Microsoft Research. 2013.
- [IP.8] S. Kane. Next generation accessible touch interfaces. Smith-Kettlewell Eye Research Institute. 2013.
- [IP.6] S. Kane. Gesture-based user interfaces for people with disabilities. IRIT (Toulouse, France). 2012.
- [IP.5] S. Kane. Prototyping and designing new assistive technologies for people with disabilities. Agency for Healthcare Research and Quality (AHRQ) Annual Conference. 2012.
- [IP.4] S. Kane. Understanding and creating usable touch interactions for blind people. HCIL Lab Talk, University of Maryland. 2012.
- [IP.3] S. Kane and K. Shinohara. OneView: Enabling collaboration between blind and sighted students using tablet computers. Microsoft Research Faculty Summit. 2010.
- [IP.2] S. Kane. Supporting independent navigation using commodity mobile phones. Technology and Disability in the Developing World Conference, University of Washington. 2009.
- [IP.1] S. Kane and M. Eisenberg. New tools for web credibility. Microsoft Research. 2007.

Technical Reports

- [T.6] S. Kane, K. Shinohara, and J. Wobbrock. *OneView: Enabling collaboration between blind and sighted students. Technical report, University of Colorado.* Tech. rep. 2015.
- [T.5] S. Kane and C. Galbraith. *Design guidelines for creating voting technology for adults with aphasia.* Tech. rep. Innovation Technology and Innovation Foundation Working Paper #006., 2013.
- [T.4] S. Kane. *Sketch-based input and evaluation in an online web-based learning environment.* Tech. rep. Master's thesis, University of Massachusetts, 2005.
- [T.3] S. Kane. *Developing modular multi-user environments with Carnival.* Tech. rep. Undergraduate honors thesis, University of Massachusetts, 2003.
- [T.2] T. Rath, S. Kane, A. Lehman, E. Partridge, and R. Manmatha. *Indexing for a digital library of George Washington's manuscripts: A study of word matching techniques.* Tech. rep. CIIR technical report MM-36, University of Massachusetts, 2002.
- [T.1] S. Kane, A. Lehman, and E. Partridge. *Indexing George Washington's handwritten manuscripts.* Tech. rep. CIIR technical report MM-34, University of Massachusetts, 2001.

Honors and Awards

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- **SIGACCESS Paper Impact Award** (2019, 2025) — Received 2 awards given to “to the authors of an ASSETS conference paper that has had a significant impact on computing and information technology that addresses the needs of persons with disabilities”, for the papers *Slide Rule: Making Mobile Touch Screens Accessible to Blind People Using Multi-Touch Interaction Techniques* and *Freedom to Roam: A Study of Mobile Device Adoption and Accessibility for People With Visual and Motor Disabilities*

- **ACM Distinguished Member** (2023)
- **JLS Paper of the Year Award** (2023) — For the paper *Reconfiguring science education through caring human inquiry and design with pets*.
- **CU Boulder: Broadening Opportunity Through Leadership and Diversity** (2018–2019) — Received fellowship from CU Boulder's College of Engineering and Applied Science to explore ways to integrate universal design into engineering education.
- **CU Boulder: Universal Design Fellowship** (2016–2017) — Member of the inaugural cohort of CU Boulder Universal Design Fellows.
- **University of Washington iSchool: Distinguished Alumni Award** (2016) — Received Distinguished Alumni Award from the University of Washington Information School.
- **UMass Amherst Computer Science: Outstanding Achievement Award** (2016) — Received *Outstanding Achievement Award by a Young Alum* from the UMass Amherst College of Information and Computer Sciences.
- **Sloan Research Fellowship** (2016) — Two-year fellowship awarded yearly to early career researchers “in recognition of distinguished performance and a unique potential to make substantial contributions to their field.”
- **UMBC Academic Innovation Fellowship** (2014) — Awarded UMBC Academic Innovation Fellowship from the UMBC Hrabowski Innovation Fund for project *Exploring Opportunities and Challenges for Wearable Computing in Classroom Settings*.
- **Google Lime Fellowship for Students with Disabilities** (2010) — Fellowship provided to outstanding students with disabilities in the field of computing.
- **Imagine Cup Touch and Tablet Accessibility Award, Finalist** (2010) — Finalist for the Touch and Tablet Accessibility Award for the project *OneView: Enabling Collaboration between Blind and Sighted Students Using Tablet Computers*.
- **NISH National Scholar Award for Workplace Innovation and Design, Hon. Mention** (2009) — Received honorable mention for the NISH National Scholar Award for the submission *Fully Accessible Touch Screens for the Blind and Visually Impaired*.
- **Most Innovative Award: UW Science and Engineering Business Association** (2008) — Received Most Innovative Award at UW SEBA Science and Technology Showcase for project *Slide Rule: Eyes-Free Mobile Phone Applications for Everyone*.
- **Graduate Student Top Scholar Award: University of Washington** (2005) — Financial support provided to top applicants to the University of Washington Graduate School.
- **Bay State Fellowship, University of Massachusetts** (2003–2005) — Assistantship presented by the Department of Computer Science at the University of Massachusetts in recognition of undergraduate academic achievement.
- **ACM SIGCSE Undergraduate Student Research Competition, Finalist** (2002) — Awarded for the poster *Indexing George Washington's Handwritten Manuscripts: A Study of Word Matching Techniques*.
- **National Science Foundation Research Experience for Undergraduates Fellowship** (2001) — NSF REU research with R. Manmatha and James Allan at the Center for Intelligent Information Retrieval at the University of Massachusetts.

Paper Awards

- **Best Paper Award Nominee: ASSETS Conference** (2020) — Received Best Paper Award nomination for the paper *Sense and Accessibility: Understanding people with physical disabilities' experiences with sensing systems*.
- **Best Paper Honorable Mention: ASSETS Conference** (2019) — Received Best Paper Award nomination for the paper *RoboGraphics: Using Mobile Robots to Create Dynamic Tactile Graphics*.

- **Honorable Mention: CHI Conference** (2019) — Received Honorable Mention (top 5% of submissions) for the paper *StoryBlocks: A Tangible Programming Game to Create Accessible Audio Stories*.
- **Honorable Mention: CHI Conference** (2017) — Received Honorable Mention (top 5% of submissions) for the paper *Toward Everyday Gaze Input: Accuracy and Precision of Eye Tracking and Implications for Design*.
- **Best Paper Award: CHI Conference** (2016) — Received Best Paper Award (top 1% of submissions) for the paper *Smart Touch: Improving Touch Accuracy for People with Motor Impairments with Template Matching*.
- **Best Paper Award: CHI Conference** (2015) — Received Best Paper Award (top 1% of submissions) for the paper *Sharing is Caring: Assistive Technology Designs on Thingiverse*.
- **Honorable Mention: CHI Conference** (2014) — Received Honorable Mention (top 5% of submissions) for the paper *Wearables and Chairables: Inclusive Design of Mobile Input and Output Devices for Power Wheelchair Users*.
- **Best Workshop Paper: IDC Conference** (2013) — Received Best Workshop Paper Award for the paper *Making “Making” Accessible*.
- **Best Paper Award: CHI Conference** (2011) — Received Best Paper Award (top 1% of submissions) for the paper *Usable Gestures for Blind People: Understanding Preference and Performance*.
- **Best Poster Award: iConference** (2009) — Received Best Poster Award for the poster *Improving Mobile Phone Accessibility with Adaptive User Interfaces*.
- **Best Paper Award: MobileHCI Conference** (2008) — Received Best Paper Award (top 2% of submissions) for the paper *Getting off the Treadmill: Evaluating Walking User Interfaces for Mobile Devices in Public Spaces*.
- **Outstanding Poster Award: ITICSE Conference** (2007) — Received Outstanding Poster Award for the poster *Engaging Student Web Programmers as Inclusive Designers*.

Research Funding and Grants

- **CAREER: A New Interaction Model for Eyes-Free Exploration of Touch Screens**
National Science Foundation, Award #IIS-1652907 · 3/1/2017–2/28/2021
Shaun Kane (PI) · \$546,610
- **CHS: Small: User Interfaces for Improving Collaboration Between Blind and Sighted People**
National Science Foundation, Award #IIS-1619384 · 9/1/2016–8/31/2019
Shaun Kane (PI) · \$499,997
- **Sloan Research Fellowship**
Alfred P. Sloan Fellowship · 3/1/2016–2/28/2018
Shaun Kane (PI) · \$55,000
- **CU Innovative Seed Grant: Increasing the Accessibility of Interactive Physics Simulations Using Actuated Tangible Interactions**
University of Colorado · 3/1/2016–12/31/2017
Shaun Kane (PI) · \$50,000
- **Promoting Strategic Research on Inclusive Access to Rich Online Content and Services**
Computing Community Consortium · 9/1/2015–12/31/2015
Clayton Lewis (PI) and Shaun Kane (Co-PI) · \$16,000
- **MRI: Acquisition of a 3D Object and Motion Capture System**
National Science Foundation, Award #CNS-1428204 · 9/1/2014–8/31/2017
Marc Olano (PI), Shaun Kane (Co-PI); Amy Hurst, Daniel Bailey, Earle Ellis (Co-PIs) · \$175,195
- **Exploring Opportunities and Challenges for Wearable Computing in Classroom Settings**
UMBC Hrabowski Innovation Fund · 1/1/2014–12/31/2014

Shaun Kane (PI) · \$14,000

- **EAGER: Understanding Barriers to Workplace Collaboration for People with Visual Impairments**
National Science Foundation, Award #IIS-1353312 · 9/15/2013–8/31/2015
Shaun Kane (PI) · \$164,982
- **Exploring Ultra-Mobile Navigation Aids for Blind People**
Toyota Engineering and Manufacturing North America · 1/1/2013–8/31/2014
Amy Hurst and Shaun Kane (Joint-PI w/ equal responsibilities) · \$380,753 (SK share 50%, \$22,399 to CU Boulder)
- **Wheeltop Interaction: Full-Body Gesture Control for Power Wheelchair Users**
Microsoft Software Engineering and Innovation Foundation · 9/1/2013–8/31/2014
Amy Hurst and Shaun Kane (Joint-PI w/ equal responsibilities) · \$25,000 (SK share 50%)
- **Accessible Touch Interfaces for Power Wheelchair and Scooter Users**
Nokia University Cooperation Funding · 9/1/2013–8/31/2014
Amy Hurst and Shaun Kane (Joint-PI w/ equal responsibilities) · \$11,450 (SK share 50%)
- **Full-Body Gesture Control for Power Wheelchair Users**
UMBC Special Research Assistantship/Initiative Support · 9/1/2013–8/31/2014
Amy Hurst and Shaun Kane (Joint-PI w/ equal responsibilities) · \$20,000 (SK share 50%)
- **VotingVoice: A Mobile Voters' Guide for People with Aphasia**
Information Technology and Innovation Foundation · 9/1/2012–8/31/2014
Shaun Kane (PI) · \$102,434
- **Audio-Haptic Feedback for Teaching Blind People to Use Touch Screens**
Google Research Award · 9/1/2012–8/31/2013
Shaun Kane (PI) · \$48,176
- **Accessible Touch Screen-Based User Interfaces for Mobile Devices**
Foundation for Science and Disability · 1/1/2010–12/31/2010
Shaun Kane (PI) · \$1,000

Teaching

Classes Taught

University of Colorado Boulder

- CSCI 4849/5849: Input, Interaction, and Accessibility (Spring 2018, Spring 2019, Fall 2020)
- CSCI 4830/7000: Inclusive Design and Assistive Technology (Spring 2017)
- CSCI 3002: Human-Centered Computing Foundations (Fall 2015, 2018, 2019, 2020)
- CSCI 4830/7000: Physical and Tangible Computing (Spring 2015)
- CSCI 5839: User-Centered Design and Development I (Fall 2014, 2016, 2017, 2019)

University of Maryland Baltimore County

- HCC 729: Human-Centered Design (Spring 2014)
- IS 403: User Interface Design (Fall 2013)
- IS 800: Mobile Human-Computer Interaction (Spring 2013)
- IS 760: Human-Computer Interaction (Fall 2011, Fall 2012)
- IS 387: Information Architecture for the World Wide Web (Spring 2012)

University of Washington

- INFO 463: Input and Interaction (Spring 2010, Spring 2011)

- INFO 344: Web Tools and Development (Summer 2006)

University of Massachusetts

- CS 121: Introduction to Problem Solving with Computers (Summer 2005)

Students Supervised

Ph.D. Advisees

- Darren Guinness, Ph.D. in Computer Science (2015–2022)
- Varsha Koushik, Ph.D. in Computer Science (2017–2022)
- Vinitha Gadiraju, Ph.D. in Computer Science (2018–2023)
- Halley P. Profita, co-advised with Nikolaus Correll, Ph.D. in Computer Science (2014–2017)
- Michele A. Williams, co-advised with Amy Hurst, Ph.D. in Human-Centered Computing at UMBC (2012–2015)

Ph.D. Supervisory Committee

- Junhan Kong, Ph.D. in Information Science, University of Washington (2025–)
- Mingyuan Zhong, Ph.D. in Information Science, University of Washington (2025–)
- Franklin Mingzhe Li, Ph.D. in Human-Computer Interaction, Carnegie Mellon University (2024–)
- Yichun Zhao, Ph.D. in Computer Science, University of Victoria (2024–)
- Guarav Jain, Ph.D. in Computer Science, Columbia University (2025–)
- Venkatesh Potluri, Ph.D. in Computer Science, University of Washington (2023–2024)
- Abir Saha, Ph.D. in Technology and Social Behavior, Northwestern University (2022–2024)
- Abigale Stangl, Ph.D. in ATLAS, University of Colorado Boulder (2015–2019)
- Redhwan Nour, Ph.D. in Computer Science, University of Colorado Boulder (2015–2017)
- Khalid Alharbi, Ph.D. in Computer Science, University of Colorado Boulder (2015–2016)
- Lise A. St. Denis, Ph.D. in Computer Science, University of Colorado Boulder (2014–2016)
- Markus Funk, Ph.D. in Human-Computer Interaction, University of Stuttgart (2016)
- Karl Wiegand, Ph.D. in Computer Science, Northeastern University (2012–2014)
- Huimin Qian, Ph.D. in Information Systems, UMBC (2011–2014)
- Shaojian Zhu, Ph.D. in Human-Centered Computing, UMBC (2012–2014)

Postdoctoral Fellows

- Stacy M. Branham, UMBC Information Systems (2013–2014). Now at UC Irvine.

Master's Thesis Advisees

- William Payne, M.S. in Computer Science, University of Colorado Boulder (2014–2016)
- Erin Duggan, co-advised with Tom Yeh, M.S. in Computer Science, University of Colorado Boulder (2014–2015)
- Ankita, M.S. in Computer Science, UMBC (2013–2014)

Master's Thesis Supervisory Committee

- Esther Vasiete, M.S. in Computer Science, University of Colorado Boulder (2014–2015)
- Jasmine Tobias, M.S. in Human-Centered Computing, UMBC (2013–2014)
- Ted O'Meara, M.S. in Human-Centered Computing, UMBC (2011)

Undergraduate Research Advisees

- Ariel Riggan, B.S. in Technology, Arts, and Media, University of Colorado. Discovery Learning Apprentice (2016–2017)
- Jesus Ortiz Tovar, B.S. in Computer Science, University of Colorado. Discovery Learning Apprentice (2016–2017)
- Christine Samson, B.S. in Computer Science, University of Colorado. Discovery Learning Apprentice (2015–2017)
- Sigrunn Sky, B.S. in Computer Science, University of Colorado. Research Assistant (2016)
- Laura Matuszewska, B.S. in Computer Science, University of Colorado. Discovery Learning Apprentice (2015)
- Lauren Gaber, B.S. in Computer Science, University of Michigan. AccessComputing Research Assistant (2015)
- Caroline Galbraith, B.S. in Information Systems, UMBC. Undergraduate Research Assistant (2012–2014)
- Beatrice Garcia, B.S. in Information Systems, UMBC. Undergraduate Research Assistant (2013–2014)
- Manpreet Suri, B.S. in Information Systems, UMBC. McNair Fellowship (2012)
- Kyle Althoff, B.S. in Information Systems, UMBC. Research Assistant (2012)
- Alec Pulianas, B.S. in Information Systems, UMBC. Research Assistant (2011–2012)
- Nicole Torcolini, Visiting Scholar, University of Washington. AccessComputing Research Assistant (2010)
- Tien Nguyen, M.S. in Information Management, University of Washington. Graduate Research Assistant (2010)
- Rishi Talwar, B.S. in Informatics, University of Washington. Directed Research Internship. (2010)
- Josh Scotland, B.S. in Computer Science, University of Washington. Undergraduate Research Assistant (2010)

Service Activities

Service to the Academic Community

Conference Chair

- General Chair, ASSETS Conference (2025–)

Journal Associate Editor

- ACM Transactions on Accessibility (2013–)
- International Journal of Human-Computer Studies (2011–2012)

Program Committees

- Program Co-Chair, ASSETS 2019 Conference
- Program Co-Chair, HCIC 2016 Conference
- Member, ACM Conference on Human Factors in Computing Systems (CHI 2012, 2013, 2015, 2017, 2018)
- Member, ACM Symposium on User Interface Software and Technology (UIST 2012, 2016, 2017)
- Member, ACM Conference on Computers and Accessibility (ASSETS 2012–2024)
- Member, ACM Web for All Conference (W4A 2015, 2018, 2019)
- Member, ACM Conference on Interaction Design and Children (IDC 2014)
- Member, ACM Conference on Intelligent User Interfaces (IUI 2009–2011)

Conference Committees

- Doctoral Consortium Co-Chair, ASSETS 2017
- Student Research Competition Judge, ASSETS 2012, ASSETS 2016
- Posters and Demonstrations Co-Chair, ASSETS 2015
- Accessibility Co-Chair, CHI 2016–2017
- Accessibility Co-Chair, CSCW 2014
- Doctoral School Panelist, ISWC 2014
- Doctoral Consortium Panelist, ASSETS 2012

Professional Organizations

- ACM SIGACCESS Member-at-Large (2016–2018)

Grant Referee

- National Science Foundation (2012, 2013, 2014, 2017, 2019)

Outreach

Advisory Boards

- Advisory board member, UW CREATE Center (2025–)
- Advisory board member, NYU Ability Project (2025–)
- Advisory board member, DIAGRAM Center (2017–2022)
- AccessComputing partner (2014–)

Outreach Events

- Computing Research Association Underrepresented Minority and Disability Graduate Student Cohort Mentor (2018, 2019, 2023)
- Diverse Learners Awareness Week, University of Colorado Boulder (2017)
- Colorado Center for the Blind STEM Education Workshop (2014–2015)
- National Federation of the Blind Youth Slam (2011, 2013, 2017)